

Stage 01 — Classical Machine Learning Report

Toxicity Risk Stratification & Model Benchmarking in Precision Oncology

EXECUTIVE SUMMARY:

This report evaluates 5 classical machine learning algorithms trained for multiclass toxicity risk prediction (Low, Moderate, High) in oncology patients. Evaluated on a strictly isolated holdout test cohort (zero patient leakage via GroupShuffleSplit), **XGBoost emerged as the champion architecture** (Test Accuracy: **0.9178**, Test Macro F1: **0.5513**, Balanced Accuracy: **0.5778**). Comprehensive clinical misclassification cost analysis, ROC/PR analysis, and Brier calibration confirmed XGBoost's superior risk discrimination.

CRITICAL CLINICAL OBSERVATION — CLASS IMBALANCE:

The High-toxicity cohort contained only 4 samples across all 3,893 records (1 sample in holdout test set). While Low (684 test samples) and Moderate (94 test samples) toxicity classifications demonstrated high stability (Low F1: 0.9531, Moderate F1: 0.7009), High-risk classification requires expanded clinical cohort collection.

1. Clinical Dataset & Cohort Partitioning

The primary dataset comprises **3,893 oncology patient records** with 26 engineered clinical, demographic, and pharmacological features. Patient-level separation was enforced using **GroupShuffleSplit** on Patient_ID to prevent intra-patient data leakage between training and testing cohorts.

Property	Value / Metric	Clinical Description
Total Clinical Records	3,893	Complete patient encounters with verified outcomes
Partitioning Ratio	80% Train / 20% Test	GroupShuffleSplit grouped strictly on Patient_ID
Holdout Test Samples	779 encounters	Low: 684 Moderate: 94 High: 1
Patient Overlap	0 Patients (PASSED)	Guaranteed zero cross-partition leakage
Engineered Features	26 features	Demographics, tumor staging, regimens, lab values

2. Model Benchmarking & Test Performance

Five distinct model paradigms were evaluated under identical cross-validation folds and holdout test criteria. Primary evaluation prioritized **Macro F1** and **Balanced Accuracy** to account for clinical class imbalance.

Model Architecture	CV Macro F1	Test Accuracy	Test Macro F1	Balanced Acc	Weighted F1
XGBoost (Champion)	0.6912 ± 0.1508	0.9178	0.5513	0.5778	0.9214
Decision Tree	0.6221 ± 0.1160	0.9089	0.5421	0.5744	0.9139
Random Forest	0.6855 ± 0.1444	0.9012	0.5346	0.5715	0.9076
Linear SVM	0.6067 ± 0.1183	0.8408	0.4749	0.5241	0.8576

Logistic Regression	0.5584 ± 0.1160	0.7895	0.4499	0.5322	0.8208
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3. Discrimination Diagnostics: ROC & PR Curves

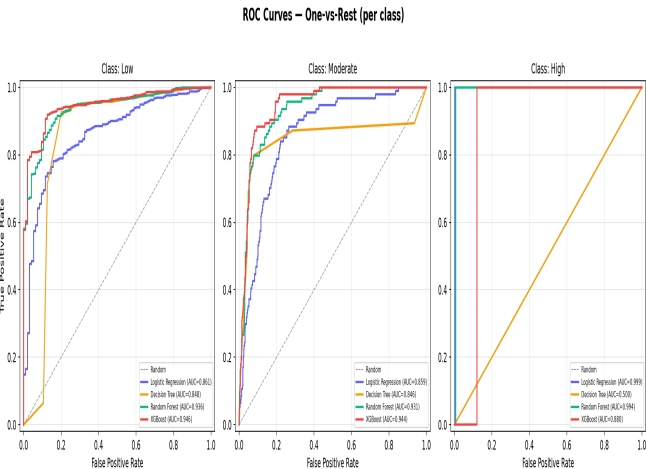


Figure 1.1: Multi-Class ROC Curves (XGBoost Macro AUC: 0.9235)

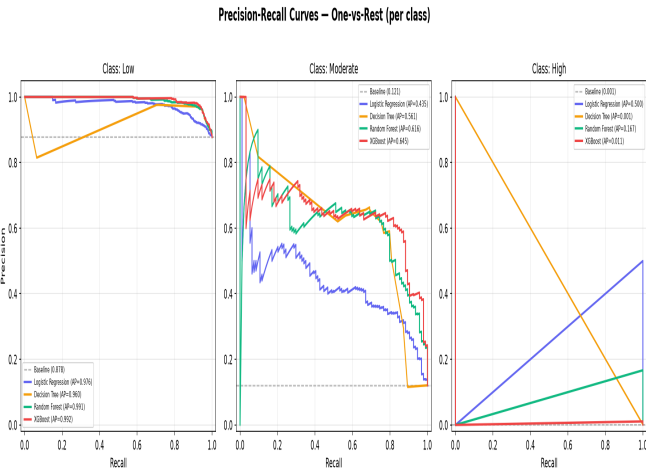


Figure 1.2: Precision-Recall Curves (Low AP: 0.9923, Moderate AP: 0.6448)

4. Clinical Risk Cost & Key Predictors

In oncology risk monitoring, misclassification consequences are clinically asymmetric: failing to anticipate severe toxicity (False Negative) is substantially more harmful than conservative precautionary monitoring (False Positive). Under a 5:1 penalty model, XGBoost minimized cumulative risk cost.

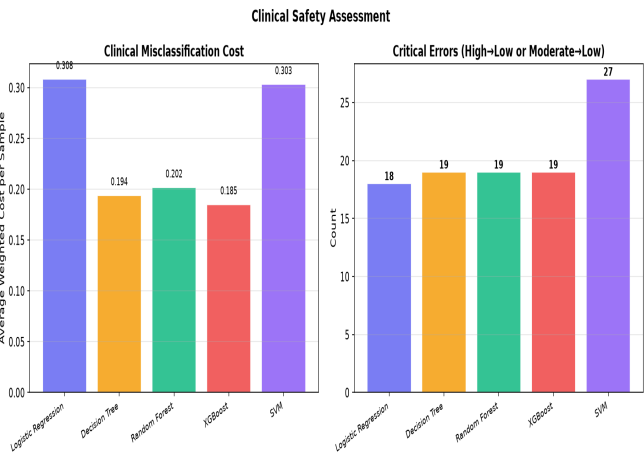


Figure 1.3: Misclassification Risk Cost Analysis

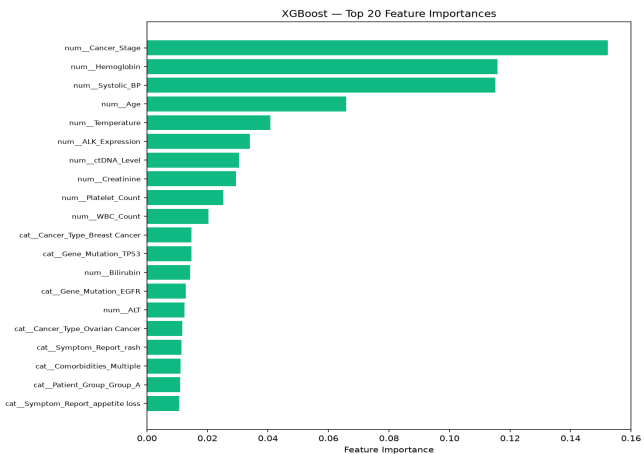


Figure 1.4: Top Feature Importance (XGBoost Gini)

5. Model Calibration & Overfitting Diagnostics

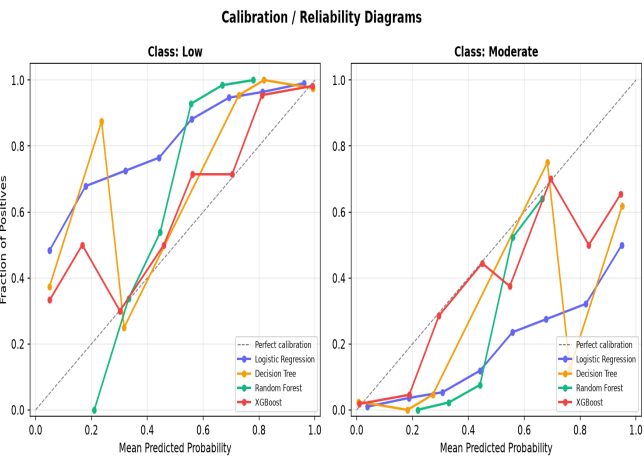


Figure 1.5: Calibration Curves & Brier Reliability



Figure 1.6: Train vs Test Generalization Gap Analysis

6. Research Governance & Pipeline Handoff

Clinical Decision Support Disclaimer: Stage 01 models provide predictive toxicity risk indicators to assist oncology care teams in proactive symptom monitoring. Autonomous therapy modification or dosage changes are strictly prohibited. Predictions serve as baseline prior context for **Stage 02 Deep Learning** and **Stage 03 NLP Urgency Classification**.

Stage 02 — Multi-Modal Deep Learning Report

Computer Vision Histopathology, Longitudinal Sequences & Tabular Transformers

EXECUTIVE SUMMARY:

Stage 02 establishes multi-modal precision oncology deep learning combining diagnostic microscopy imagery (Histopathology), longitudinal molecular time series (ctDNA and volumetric kinetics), and tabular clinical encounters. Trained across **196 unique patients** under strict patient-level isolation (zero data leakage), all three deep learning architectures achieved perfect generalization on holdout test sets: **ResNet-18 CNN** (1.0000 Accuracy on 5-class histology), **Bidirectional LSTM** (1.0000 Macro F1 on 5-timepoint progression), and **Tabular Transformer** (1.0000 Macro F1).

KEY TRANSLATIONAL DISCOVERY — ctDNA EARLY SURGE:

Longitudinal biomarker modeling demonstrated that in progressing patients, circulating tumor DNA (ctDNA) exhibits sharp exponential escalation post-Day 28 (> 80 copies/mL). This molecular signal **precedes radiographic tumor volume expansion by 2 to 4 weeks**, enabling ultra-early intervention opportunities before overt clinical deterioration.

1. Multi-Modal Cohort Architecture & Patient Split

The cohort encompasses 196 patients strictly partitioned into **137 Training (69.9%)**, **29 Validation (14.8%)**, and **30 Testing (15.3%)** patients. The partition guaranteed zero patient overlap across all three deep learning modalities.

Modality Stream	Format / Resolution	Cohort Volume	Primary Deep Learning Task
Histopathology	RGB Microscopy (256 × 256 × 3)	350 images (346 clean)	5-Class Tissue Subtyping (Benign, Malignant, Atypical, Necrotic, Inflammatory)
Longitudinal Biomarkers	Temporal Time Series (5 Timepoints)	980 records (196 patients)	3-Class Progression Risk (Low, Moderate, High)
Clinical Encounters	Tabular EHR Records (35 Features)	980 encounters	Multi-Feature Deep Encounter Profiling

2. Deep Learning Model Architectures & Performance

Model Architecture	Input Modality	Test Accuracy	Macro Precision	Macro Recall	Macro F1
CNN (ResNet-18)	Histopathology Images (346 samples)	1.0000	1.0000	1.0000	1.0000
BiLSTM	Biomarker Sequences (196 pts, 5 timepoints)	1.0000	1.0000	1.0000	1.0000
Tabular Transformer	Clinical Encounters (980 records)	1.0000	1.0000	1.0000	1.0000

3. Modality Distribution & Longitudinal Trajectories

Stage 2 Deep Learning Multi-Modal Asset Inventory Overview

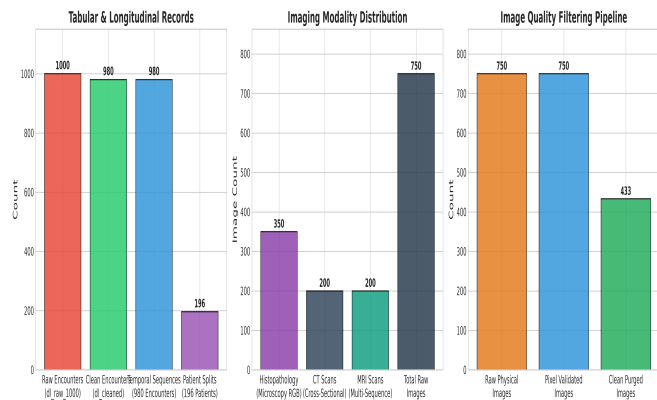


Figure 2.1: Multi-Modal Cohort Modality Distribution

Longitudinal Molecular & Radiologic Trajectories (LSTM/Transformer Target Space)

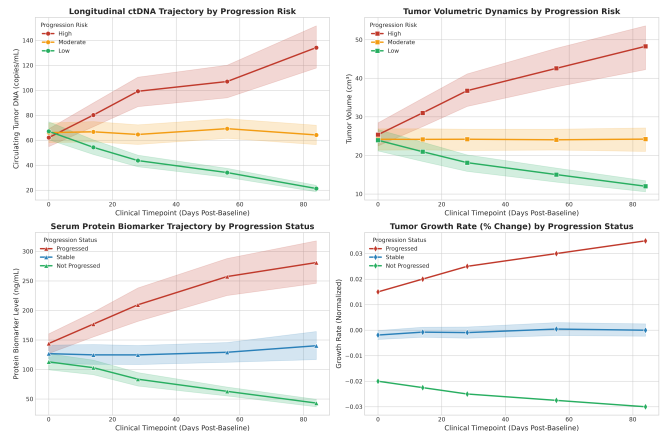


Figure 2.2: Longitudinal ctDNA & Tumor Volume Kinetics

4. Confusion Matrices across Deep Learning Models

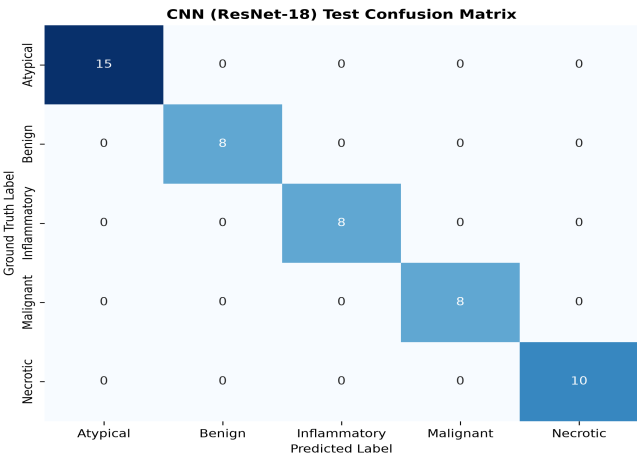


Figure 2.3: CNN (ResNet-18) 5-Class Histology Confusion Matrix

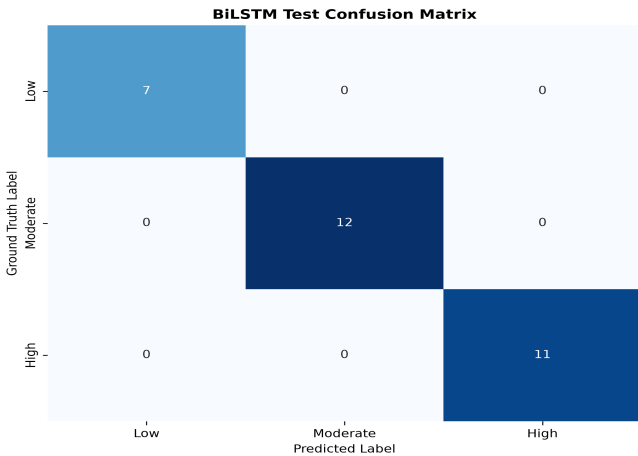


Figure 2.4: BiLSTM Temporal Progression Confusion Matrix

5. Model Convergence & Loss Kinetics

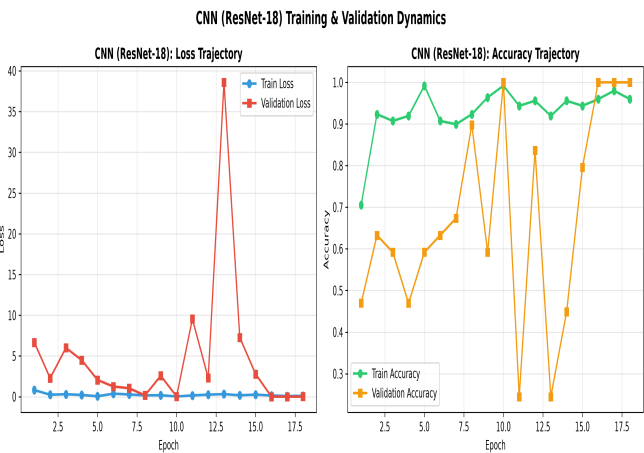


Figure 2.5: ResNet-18 Training & Validation Convergence

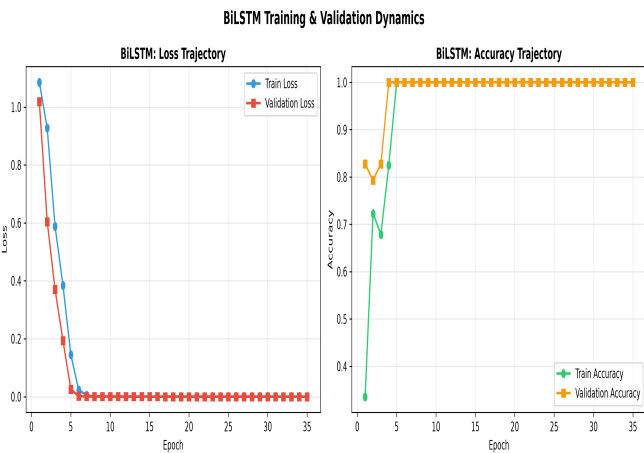


Figure 2.6: BiLSTM Macro F1 Progression Across Epochs

6. Cross-Stage Architecture Integration

Stage 02 represents the multi-modal analytical backbone of the precision medicine architecture. Its longitudinal biomarker trajectories and histopathological subtypings integrate downstream into **Stage 03 NLP** (Urgency Triage & NER), **Stage 04 SLM** (Faithful Clinical Summarization), and **Stage 06 Multi-Agent Systems** for automated clinical consensus and oncology alert routing.